

Technology Stack

Date 03 October 2026
Team ID COMICCRAFT-TEAM
Project Name ComicCrafter AI – AI-Powered Comic Strip Generator
Maximum Marks 2 Marks

Technology Stack Details

S.No	Architecture	Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend	Web UI	Streamlit	Provides the interactive prompt, style selector, progress messages, preview and download controls.
2	Application	Programming Language	Python	Implements the complete generation and processing workflow.
3	AI Text	Panel generation	Google Generative AI – Gemini 1.5 Flash	Creates six structured panel descriptions and dialogue from the story.
4	AI Image	Panel artwork	ClipDrop Text-to-Image API	Generates one visual image for each panel.
5	Image Processing	Comic assembly	Pillow (PIL)	Adds borders/text areas, wraps dialogue and combines panels into a 3×2 grid.
6	Document Output	PDF	ReportLab	Creates the downloadable A4 PDF from the final comic strip.
7	Configuration	Secrets	python-dotenv	Loads GEMINI_API_KEY and CLIPDROP_API_KEY from environment variables.
8	HTTP	API calls	requests	Sends requests to the ClipDrop image generation service.